WORKED SOLUTION

STEP II 2018, Question 11

Mechanics • Friction and Moments

STEP II 2018, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q11

The key idea

Three equations pin down the three unknown contact forces: Newton's second law horizontally, no vertical acceleration, and the given condition that the moments about the centre of mass balance. Once both normal reactions are written in terms of the acceleration a, each of the two failures becomes a threshold on a. The front wheel lifts when its reaction falls to zero, and the rear wheel slips when the friction it needs exceeds μ times its reaction. Whichever threshold is lower happens first.

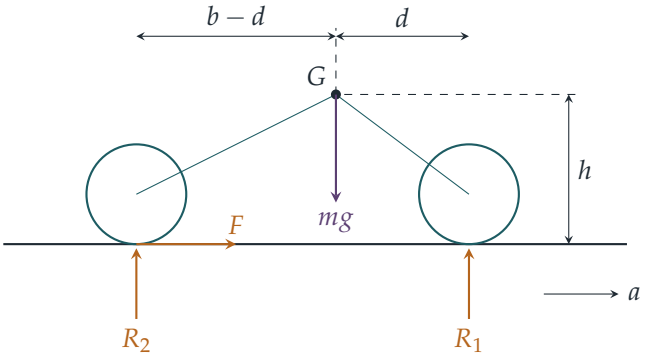
Solution

The model and the forces

Treat the motorbike as a rigid body moving in a straight line with acceleration $a \geq 0$ to the right. The road is horizontal, so the centre of mass G has no vertical acceleration and the bike does not rotate. The forces are:

- the weight mg acting down at G;
- the normal reaction R_1 at the front wheel (which is smooth, so there is no friction there);
- the normal reaction R_2 and the friction F at the rear wheel. The engine drives the rear wheel, so F acts forwards and is the only horizontal force.

The front contact point is a horizontal distance d ahead of G and the rear one is $b - d$ behind it. Both are a distance h below G.



The reactions in terms of a

Horizontally and vertically,

$$F = ma, \qquad R_1 + R_2 = mg.$$

Now take moments about G, with anticlockwise positive (the sense that would lift the front wheel). The weight has no moment about G. R_1, acting up a distance d ahead of G, has anticlockwise moment $R_1 d$. R_2, acting up a distance $b - d$ behind G, has clockwise moment $R_2(b - d)$. And F acts forwards along a line a distance h below G, so it has anticlockwise moment Fh. Setting the total to zero:

$$R_1 d + Fh = R_2(b - d).$$

Substitute $F = ma$ and $R_1 = mg - R_2$:

$$mgd - R_2 d + mah = R_2(b - d) \quad \Longrightarrow \quad R_2 = \frac{m(gd + ah)}{b},$$

and then

$$R_1 = mg - R_2 = \frac{m\big((b - d)g - ah\big)}{b}.$$

As a increases, weight transfers from the front wheel to the rear: R_1 decreases and R_2 increases.

When does each failure happen?

Front wheel. The front wheel stays on the road while $R_1 \geq 0$, that is while

$$a \leq \frac{(b - d)g}{h} =: a_{\text{lift}}.$$

Rear wheel. The rear wheel does not slip while $F \leq \mu R_2$, that is while

$$ma \leq \frac{\mu m(gd + ah)}{b} \iff a(b - \mu h) \leq \mu dg.$$

Ask yourself Why can we divide by $b - \mu h$ without reversing the inequality?

This is exactly what the given condition $\mu < b/h$ is for: it says $b - \mu h > 0$. So the rear wheel grips while

$$a \leq \frac{\mu d g}{b - \mu h} =: a_{\text{slip}}.$$

(Without that condition the friction available would grow at least as fast as the friction needed, and the rear wheel would never slip.)

Both thresholds are positive, and as a increases from zero the rear wheel slips first exactly when $a_{\text{slip}} < a_{\text{lift}}$. Since $b - \mu h > 0$ and $h > 0$ we can multiply out:

$$\begin{aligned} \frac{\mu d g}{b - \mu h} < \frac{(b-d)g}{h} &\iff \mu d h < (b-d)(b - \mu h) \\ &\iff \mu d h + \mu h (b-d) < b(b-d) \\ &\iff \mu h b < b(b-d) \\ &\iff \mu < \frac{b-d}{h}, \end{aligned}$$

dividing by $bh > 0$ at the last step. So the rear wheel slips before the front wheel loses contact if $(*)$ holds, as required. (In fact the chain shows the two statements are equivalent.)

Maximum acceleration when $(*)$ holds

If $(*)$ holds then $a_{\text{slip}} < a_{\text{lift}}$. Any acceleration up to a_{slip} satisfies both conditions, so the front wheel stays down and the rear wheel grips. Any larger acceleration would need $F > \mu R_2$. So the maximum acceleration without slipping is

$$a_{\max} = \frac{\mu d g}{b - \mu h},$$

as required.

When $(*)$ does not hold

Now $\mu \geq \dfrac{b-d}{h}$, so the same chain of equivalences (with $\geq$ in place of $<$) gives $a_{\text{lift}} \leq a_{\text{slip}}$. Every acceleration up to a_{lift} therefore keeps $R_1 \geq 0$ and also satisfies $a \leq a_{\text{slip}}$, so the rear wheel does not slip. Beyond a_{lift} we would need $R_1 < 0$, which is impossible while the front wheel touches the road. So the maximum acceleration with the front wheel in contact is

$$a_{\max} = \frac{(b-d)g}{h}.$$

At this acceleration $R_1 = 0$: the bike is just on the point of a wheelie, with all its weight on the rear wheel.

Checks

- At $a = 0$: $R_2 = mgd/b$ and $R_1 = mg(b-d)/b$, the usual static split by the lever rule. The wheel nearer G carries more weight.
- In the boundary case $\mu = (b-d)/h$ we have $b - \mu h = d$, so $a_{\text{slip}} = \mu g = (b-d)g/h = a_{\text{lift}}$. Both answers agree, as they must.
- A numerical example: $b = 1.5\,\text{m}$, $d = 0.75\,\text{m}$, $h = 0.6\,\text{m}$, $\mu = 0.8$, $g = 9.8\,\text{m}\,\text{s}^{-2}$. Here $(b-d)/h = 1.25 > \mu$, and $a_{\text{slip}} = 5.88/1.02 \approx 5.76\,\text{m}\,\text{s}^{-2}$, well below $a_{\text{lift}} = 12.25\,\text{m}\,\text{s}^{-2}$. A computer check over many random bikes, stepping a upwards and recording which condition fails first, agreed with both formulae.

Common slips

- Getting the sign of the friction moment wrong. Forward friction below G tends to lift the front, which is why acceleration unloads the front wheel.
- Dividing by $b - \mu h$ without saying why it is positive. That is the only job of the condition $\mu < b/h$.
- Taking moments about a contact point and setting the total to zero. The bike is accelerating, so the moments balance about G but not about other points.

Beyond STEP

The given condition about moments is a special case of the rigid-body law that the rate of change of angular momentum about the centre of mass equals the total torque about it. Here the angular momentum is zero, since the bike does not rotate. About any other point an accelerating body acquires an extra "fictitious" torque from ma, which explains the last common slip. The same weight-transfer analysis underlies braking and traction limits in vehicle dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body dynamics carefully.

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